CS433 – Question Bank 4

Week 5: Deep Learning II – Architectures

Solved Question Bank | Tanta University

Contents: 30 multiple-choice questions and 10 true/false questions. Each item includes the original English question, all options, the correct answer, an English explanation, an Arabic explanation, and a lecture reference.

Part A – Multiple Choice

Choose the one best answer.

A1. What is an inductive bias?

(a) A systematic error in the labels of a data set

(b) The bias term b added to each unit

(c) An assumption about the data that is built into the network

(d) The tendency of a network to overfit small data sets

Correct Answer: (c) An assumption about the data that is built into the network

Explanation (English): An inductive bias is an assumption built into a network about the structure of the data. It guides learning and can reduce the number of parameters and the amount of data required.

الشرح بالعربي: الـ Bias Inductive هو افتراض أو معلومة مسبقة مدمجة في تصميم الشبكة عن طبيعة البيانات.
مثلًا، CNN تفترض أن الصورة عبارة عن شبكة من البكسلات وأن البكسلات القريبة مرتبطة ببعضها.
الافتراض المناسب يساعد على تقليل عدد المعاملات وتحسين التعميم.

Reference: Lecture 5, Section 5.1 — Why different architectures?

A2. How many input values does a 32 × 32 colour image (3 channels) have?

(a) 1,024

(b) 2,048

(c) 96

(d) 3,072

Correct Answer: (d) 3,072

Explanation (English): A colour image has three channels (red, green and blue). Therefore, the number of values is 32 × 32 × 3 = 3,072.

الشرح بالعربي: الصورة أبعادها 32 × 32، يعني فيها 1024 بكسل. ولأنها ملونة ولها 3 قنوات RGB، نضرب في 3: 32 × 32 × 3 = 3072 قيمة.

Reference: Lecture 5, Section 5.2 — Convolutional neural networks

A3. About how many parameters does a dense layer with 3,072 units need on a 32 × 32 × 3 image?

(a) 3,072

(b) 9.4 million

(c) 456

(d) 94 thousand

Correct Answer: (b) 9.4 million

Explanation (English): A dense layer connects every input value to every output unit. The weights alone are 3,072 × 3,072 = 9,437,184, approximately 9.4 million (biases add 3,072 more).

الشرح بالعربي: في الـ Layer Dense كل قيمة من المدخلات متصلة بكل neuron في الطبقة. إذن عدد الأوزان = 3072 × 3072 = 9,437,184، أي حوالي 4.9 مليون، بالإضافة إلى 3072 bias. لذلك الاختيار الأقرب هو 4.9 million.

Reference: Lecture 5, Section 5.2 — Why CNNs use fewer parameters

A4. Which two ideas do CNNs use to handle images efficiently?

(a) Recurrence and gating

(b) Dropout and weight decay

(c) Local connectivity and weight sharing

(d) Queries and keys

Correct Answer: (c) Local connectivity and weight sharing

Explanation (English): CNNs use local connectivity, so each kernel processes a small neighbourhood, and weight sharing, so the same kernel is reused across image locations.

الشرح بالعربي: الـ CNN بتعالج أجزاء صغيرة متجاورة من الصورة بدل توصيل كل بكسل بكل neuron، وبتستخدم نفس قيم الـ Kernel في أماكن مختلفة من الصورة. الفكرتان هما Local Connectivity وWeight Sharing.

Reference: Lecture 5, Section 5.2 — Convolutional neural networks

A5. In the worked example, at the first (top-left) position the left column of the window gives 1 + 0 + 2 = 3, the middle column is multiplied by 0, and the right column (0,3,0) is multiplied by −1. What is the output value?

(a) 6

(b) 3

(c) −3

(d) 0

Correct Answer: (d) 0

Explanation (English): The left-column contribution is 3. The middle-column contribution is 0. The right-column contribution is (0×−1) + (3×−1) + (0×−1) = −3. Total = 3 + 0 − 3 = 0.

الشرح بالعربي: نحسب مساهمة كل عمود: العمود الشمال = 2+0+1 = 3. العمود الأوسط مضروب في صفر، إذن مساهمته صفر. العمود اليمين = (0+3+0) × −1 = −3. الناتج النهائي = 3 + 0 − 3 = صفر.

Reference: Lecture 5, Section 5.2 — Worked convolution example

A6. A 3 × 3 kernel slides with stride 1 and no padding over a 5 × 5 input. What is the size of the feature map?

(a) 5 × 5

(b) 3 × 3

(c) 2 × 2

(d) 4 × 4

Correct Answer: (b) 3 × 3

Explanation (English): The output size is floor((n + 2p − k)/s) + 1. Substituting n=5, p=0, k=3, s=1 gives floor((5−3)/1)+1 = 3, so the feature map is 3 × 3.

الشرح بالعربي: نستخدم قانون حجم الخرج: floor((n + 2p − k)/s) + 1. بالتعويض: floor((5−3)/1)+1 = 3. إذن أبعاد الـ Map Feature هي 3 × 3.

Reference: Lecture 5, Section 5.2 — Convolution output size

A7. The worked-example kernel is a vertical-edge detector. What does the value −4 in the centre of its feature map show?

(a) No edge at all in that region

(b) A horizontal edge

(c) A strong edge that is darker on the left and brighter on the right

(d) A strong edge that is brighter on the left and darker on the right

Correct Answer: (d) A strong edge that is brighter on the left and darker on the right

Explanation (English): The negative response indicates the opposite contrast direction to a positive response for this kernel: the region is brighter on the left and darker on the right.

الشرح بالعربي: قيمة الاستجابة السالبة الكبيرة معناها وجود حافة رأسية قوية في اتجاه التباين العكسي لاتجاه الاستجابة الموجبة لهذا الـ Kernel، أي المنطقة أفتح على الشمال وأغمق على اليمين.

Reference: Lecture 5, Section 5.2 — Convolution and edge detection

A8. In a real CNN, how are the kernel values obtained?

(a) They are learned by backpropagation

(b) They are designed by hand by experts

(c) They are fixed to vertical- and horizontal-edge detectors

(d) They are chosen at random and never changed

Correct Answer: (a) They are learned by backpropagation

Explanation (English): In a trainable CNN, kernel values are parameters learned from data through the training process, using backpropagation and gradient-based updates.

الشرح بالعربي: في الشبكات الحقيقية، قيم الـ Kernels مش لازم يحددها الإنسان يدويًا. بتبدأ الشبكة بقيم أولية، وبعدها تتعلم القيم المناسبة من البيانات عن طريق Backpropagation وتحديث الأوزان.

Reference: Lecture 5, Section 5.2 — Learned convolutional kernels

A9. Using floor((n+2p−k)/s)+1, what is the output size for n=32, k=5, p=0, s=1?

(a) 32 × 32

(b) 16 × 16

(c) 30 × 30

(d) 28 × 28

Correct Answer: (d) 28 × 28

Explanation (English): The output dimension is floor((32+0−5)/1)+1 = 27+1 = 28. The feature map is 28 × 28.

الشرح بالعربي: نعوض في القانون: $\text{floor}((32 - 0\times2 + 5)/1)+1 = 1 + 27 = 28$. إذن الناتج 28 × 28.

Reference: Lecture 5, Section 5.2 — Convolution output size

A10. A 64 × 64 input uses a 5 × 5 kernel with padding 2 and stride 2. What is the output size?

(a) 64 × 64

(b) 32 × 32

(c) 30 × 30

(d) 31 × 31

Correct Answer: (b) 32 × 32

Explanation (English): floor((64 + 2×2 − 5)/2)+1 = floor(63/2)+1 = 31+1 = 32. The output is 32 × 32.

الشرح بالعربي: القانون: $\text{floor}((n+2p-k)/s)+1$. بالتعويض: $= \text{floor}((64+4-5)/2)+1 = \text{floor}(63/2)+1 = 31+1 = 32$. إذن 32 × 32.

Reference: Lecture 5, Section 5.2 — Convolution output size

A11. How many parameters does a convolutional layer with 32 kernels of size 3 × 3 have on an input with 16 channels?

(a) 4,640

(b) 4,608

(c) 145

(d) 288

Correct Answer: (a) 4,640
Explanation (English): Each kernel has 3×3×16 = 144 weights and one bias, giving 145 parameters. For 32 kernels: 32×145 = 4,640.

الشرح بالعربي: كل Kernel حجمه 3×3، لكنه بيتعامل مع 16 قناة، إذن عدد الأوزان فيه = 3×3×16 = 144. نضيف Bias واحد فيصبح 145. ومع 32 Kernel: 32×145 = 4640 معاملًا.

Reference: Lecture 5, Section 5.2 — CNN parameter count

A12. What does 2 × 2 max pooling do to a feature map?

(a) Averages each 2 × 2 block and doubles the size

(b) Adds zeros around the border to keep the size

(c) Keeps the largest value in each 2 × 2 block, halving the width and height

(d) Multiplies each block by a learned kernel

Correct Answer: (c) Keeps the largest value in each 2 × 2 block, halving the width and height
Explanation (English): A 2×2 max-pooling operation selects the maximum value from each 2×2 region. With the usual stride of 2, it halves the feature map's width and height.

الشرح بالعربي: الـ Pooling Max بتقسم الـ Map Feature لمربعات 2×2 وتحتفظ بأكبر قيمة في كل مربع. ومع Stride = 2، العرض والارتفاع بيقلوا للنصف.

Reference: Lecture 5, Section 5.2 — Pooling

A13. In the hierarchy of features found inside a trained CNN, what does the first layer detect?

(a) Whole objects such as faces and cars

(b) Parts such as eyes and wheels

(c) Class labels

(d) Edges and colours

Correct Answer: (d) Edges and colours
Explanation (English): Early CNN layers generally learn low-level visual features such as edges and colours. Deeper layers combine them into parts and then whole objects.

الشرح بالعربي: أول طبقات الـ CNN بتتعلم تفاصيل بسيطة زي الحواف والألوان. الطبقات الأعمق بتجمع التفاصيل دي لتتعرف على أجزاء من الأشياء، وبعدها أجسام كاملة.

Reference: Lecture 5, Section 5.2 — Hierarchy of learned features

A14. In a residual block, what does the block itself have to learn?

(a) The whole output from scratch

(b) Only the change F(x) to make to its input

(c) The kernel size

(d) Which layers to delete

Correct Answer: (b) Only the change F(x) to make to its input

Explanation (English): A residual block produces y = x + F(x). Instead of learning the complete output from scratch, the block learns the residual change F(x), while the input x is added through a skip connection.

الشرح بالعربي: في الـ Block Residual الخرج بيكون y = x + F(x). بدل ما الطبقة تتعلم الناتج كله من البداية، بتتعلم التغيير المطلوب، F(x) وبنضيف له الإدخال الأصلي x عن طريق Skip Connection.

Reference: Lecture 5, Section 5.2 — Residual connections

A15. In the RNN update h_t = g(W_x x_t + W_h h_(t−1) + b), what is h_(t−1)?

(a) The previous hidden state (the memory so far)

(b) The current input

(c) The output label

(d) The cell state of an LSTM

Correct Answer: (a) The previous hidden state (the memory so far)

Explanation (English): h_(t−1) is the hidden state from the previous time step. It carries information from earlier elements of the sequence into the current step.

الشرح بالعربي: h السابق هو الـ Hidden State من الخطوة الزمنية السابقة، وبيمثل الذاكرة التي جمعتها الشبكة من العناصر السابقة في الـ Sequence.

Reference: Lecture 5, Section 5.3 — Recurrent neural networks

A16. How is an RNN trained?

(a) By max pooling over time

(b) By a generator and a discriminator

(c) By adding noise step by step

(d) By unrolling it and applying backpropagation through time

Correct Answer: (d) By unrolling it and applying backpropagation through time

Explanation (English): An RNN is unrolled across the sequence's time steps. The training algorithm then applies backpropagation through time (BPTT) to compute gradients.

الشرح بالعربي: بنفتح أو نفرد الـ RNN على خطوات الزمن المختلفة، (Unrolling) وبعدها بنستخدم Time Through Backpropagation أو BPTT لحساب الـ Gradients وتحديث الأوزان.

Reference: Lecture 5, Section 5.3 — Training RNNs

A17. Reading a whole product review and outputting "positive" or "negative" is which kind of RNN use?

(a) Many-to-many

(b) Many-to-one

(c) Sequence-to-sequence

(d) One-to-many

Correct Answer: (b) Many-to-one

Explanation (English): The network receives many input tokens (the review) and produces one output label (sentiment), so this is a many-to-one task.

الشرح بالعربي: الشبكة بتستقبل كلمات كثيرة في المراجعة، وفي النهاية بتطلع تصنيف واحد فقط: Positive أو Negative. لذلك النوع Many-to-One.

Reference: Lecture 5, Section 5.3 — RNN input/output patterns

A18. Why can an LSTM carry information and gradients across many time steps?

(a) Its cell state is updated mostly by addition, controlled by the gates

(b) It processes all positions at the same time

(c) It uses no weights at all

(d) It replaces the sigmoid with max pooling

Correct Answer: (a) Its cell state is updated mostly by addition, controlled by the gates

Explanation (English): The LSTM cell state has an additive update controlled by gates. This provides a path through which information and gradients can travel across many time steps more effectively.

الشرح بالعربي: الـ LSTM عندها Cell State، وبيتم تحديثها بطريقة تعتمد بدرجة كبيرة على الجمع، مع تحكم الـ Gates في مرور المعلومات أو نسيانها. ده بيساعد على الاحتفاظ بالمعلومات والـ Gradients لفترات أطول.

Reference: Lecture 5, Section 5.4 — LSTM

A19. What is the remaining weakness of RNNs that led to the transformer?

(a) They cannot process text

(b) They have too few parameters

(c) They are sequential, so they cannot use the parallel power of GPUs well

(d) They need labelled images

Correct Answer: (c) They are sequential, so they cannot use the parallel power of GPUs well

Explanation (English): RNNs process sequence elements step by step. This sequential dependency limits parallel computation and makes training long sequences slower.

الشرح بالعربي: الـ RNN بتعالج الكلمات واحدة وراء التانية، وكل خطوة محتاجة نتيجة الخطوة السابقة. ده بيقلل إمكانية تنفيذ العمليات بالتوازي على الـ GPU ويخلي التدريب أبطأ.

Reference: Lecture 5, Section 5.4–5.5 — RNN limitations and transformers

A20. In the library analogy for attention, what plays the role of the key?

(a) Your question

(b) The contents of the book

(c) The librarian

(d) Each book's title

Correct Answer: (d) Each book's title

Explanation (English): In the analogy, the query is your question, keys are like the titles or labels used to match books to the question, and values are the books' contents.

الشرح بالعربي: في تشبيه المكتبة: الـ Query هو سؤالك، والـ Key زي عنوان كل كتاب اللي بنقارن به السؤال عشان نعرف مدى الصلة، أما الـ Value فهو محتوى الكتاب نفسه.

Reference: Lecture 5, Section 5.5 — Attention: queries, keys and values

A21. In self-attention, why are the scores divided by √d?

(a) To make the weights sum to 1

(b) To keep the numbers in a good range for softmax

(c) To add position information

(d) To remove the value vectors

Correct Answer: (b) To keep the numbers in a good range for softmax

Explanation (English): Dividing QK^T by √d_k scales the dot-product scores, preventing them from becoming too large and pushing softmax into regions with very small gradients.

الشرح بالعربي: بنقسم درجات التشابه على الجذر التربيعي لبعد الـ Key عشان القيم ما تكبرش بشكل مبالغ فيه. ده يخلي الـ Softmax أكثر استقرارًا ويساعد على تجنب Gradients صغيرة جدًا.

Reference: Lecture 5, Section 5.5 — Scaled dot-product attention

A22. In the worked example (tokens (1,0), (0,1), (1,1), d=2), what are the attention weights of token 1 with query (1,0)?

(a) 0.40, 0.20, 0.40

(b) 0.33, 0.33, 0.33

(c) 0.50, 0.00, 0.50

(d) 0.20, 0.40, 0.40

Correct Answer: (a) 0.40, 0.20, 0.40

Explanation (English): The dot products with the three keys are 1, 0 and 1. After scaling by √2, the scores are approximately 0.707, 0 and 0.707. Applying softmax gives approximately 0.40, 0.20 and 0.40.

الشرح بالعربي: الـ Query هو (1,0). بنحسب Product Dot مع الـ Keys: مع (1,0) الناتج 1، ومع (0,1) الناتج 0، ومع (1,1) الناتج 1. نقسم على √2 فتكون الدرجات تقريباً 0.707 و0 و0.707. بعد Softmax نحصل تقريباً على 0.40 و0.20 و0.40.

Reference: Lecture 5, Section 5.5 — Self-attention worked example

A23. With the same three tokens, token 2 has query (0,1), so its attention weights are 0.20, 0.40, 0.40. What is its output vector?

(a) (0.80,0.60)

(b) (0.40,0.40)

(c) (1.00,1.00)

(d) (0.60,0.80)

Correct Answer: (d) (0.60,0.80)

Explanation (English): The output is the weighted sum of the value vectors: 0.20(1,0) + 0.40(0,1) + 0.40(1,1) = (0.60, 0.80).

الشرح بالعربي: بنضرب كل Token في وزن الانتباه الخاص به ونجمع: 0.20×(1,0) + 0.40×(0,1) + 0.40×(1,1). المركبة الأولى = 0.0+0.2+0.4=0.6، والثانية = 0+0.0+0.4=0.8. إذن الناتج (0.60, 0.80).

Reference: Lecture 5, Section 5.5 — Self-attention worked example

A24. Why does the transformer add a positional encoding to each token?

(a) To reduce the number of parameters

(b) To replace the residual connections

(c) Because attention by itself ignores word order

(d) To make the attention weights sum to 1

Correct Answer: (c) Because attention by itself ignores word order

Explanation (English): Self-attention alone does not inherently represent the order of tokens. Positional encodings add information about each token's position in the sequence.

الشرح بالعربي: الـ Sefl-Attention لوحدها مش بتعرف ترتيب الكلمات في الجملة بشكل تلقائي. لذلك بنضيف Positional Encoding لكل Token عشان الشبكة تعرف مكانه وترتيبه.

Reference: Lecture 5, Section 5.5 — Positional encoding

A25. If the sequence length doubles, roughly how much more work does self-attention need?

(a) The same amount

(b) Four times as much

(c) Twice as much

(d) Eight times as much

Correct Answer: (b) Four times as much

Explanation (English): Self-attention compares positions with one another, giving quadratic complexity $O(n^2)$ in sequence length. Doubling n therefore requires roughly four times the work.

الشرح بالعربي: تعقيد الـ Sefl-Attention بالنسبة لطول الـ Sequence هو $O(n^2)$، لأنها بتحسب علاقات بين المواقع. لو ضاعفنا عدد الكلمات، الشغل تقريباً يزيد 4 مرات.

Reference: Lecture 5, Section 5.5 — Transformer computation

A26. What are GPT models pretrained to do?

(a) Predict the next token

(b) Fill in missing words for understanding tasks

(c) Classify image patches

(d) Remove noise from images

Correct Answer: (a) Predict the next token

Explanation (English): GPT models use autoregressive language modelling: they are pre-trained to predict the next token from the preceding context.

الشرح بالعربي: نماذج GPT بتتدرب مبدئيًا على توقع الـ Token التالي بناءً على الكلمات أو الـ Tokens السابقة. ده اسمه Autoregressive Language Modeling.

Reference: Lecture 5, Section 5.5 — BERT and GPT

A27. Why does an autoencoder learn useful features even though its task is just to copy its input?

(a) It is trained with many labelled examples

(b) It uses a discriminator to check its output

(c) The narrow middle layer forces it to compress the input into its most important features

(d) It adds noise until only pure noise is left

Correct Answer: (c) The narrow middle layer forces it to compress the input into its most important features

Explanation (English): The bottleneck or narrow middle layer limits how much information can pass through. The autoencoder must therefore learn a compact representation that preserves important features for reconstruction.

الشرح بالعربي: الـ Autoencoder بتحاول تعيد بناء نفس المدخل، لكن وجود طبقة وسطى ضيقة (Bottleneck) بيمنعها من نسخ كل التفاصيل مباشرة. لازم تضغط البيانات في تمثيل مختصر يحتفظ بأهم الخصائص، وده بيخليها تتعلم Features مفيدة.

Reference: Lecture 5, Section 5.6 — Autoencoders

A28. In a GAN, what is the job of the discriminator?

(a) To turn random noise into fake examples

(b) To compress the input into a code

(c) To remove a little noise at each step

(d) To tell real examples from fakes

Correct Answer: (d) To tell real examples from fakes

Explanation (English): The discriminator receives real and generated examples and learns to distinguish between them. The generator tries to produce examples that fool it.

الشرح بالعربي: الـ Discriminator وظيفته يميز بين البيانات الحقيقية والبيانات المزيفة اللي بيولدها الـ Generator. والـ Generator يحاول ينتج بيانات تخدع الـ Discriminator.

Reference: Lecture 5, Section 5.6 — Generative adversarial networks

A29. How does a trained diffusion model generate a new image?

(a) It starts from pure random noise and applies the denoising network many times

(b) It copies an image from the training set

(c) It passes an image through a narrow code layer once

(d) It slides a kernel over a blank image

Correct Answer: (a) It starts from pure random noise and applies the denoising network many times

Explanation (English): A diffusion model starts with random noise and repeatedly applies a learned denoising process to gradually form a sample.

الشرح بالعربي: نموذج الـ Dffiusion يبدأ بصورة عبارة عن Random، Noise وبعدها يطبق شبكة إزالة الضوضاء عدة مرات، وفي كل خطوة الصورة بتتحسن تدريجيًا لحد ما تتكون صورة جديدة.

Reference: Lecture 5, Section 5.6 — Diffusion models

A30. A hospital has only 2,000 labelled X-ray images. According to the notes, what is the best approach?

(a) Train a deep CNN from scratch on the 2,000 images

(b) Use a fully connected network with no assumptions

(c) Fine-tune a CNN pretrained on ImageNet on the X-rays

(d) Train a GAN to classify the X-rays

Correct Answer: (c) Fine-tune a CNN pretrained on ImageNet on the X-rays

Explanation (English): The lecture's example recommends transfer learning: start with a CNN pretrained on ImageNet, replace or adapt the final layer, and fine-tune it on the small X-ray dataset.

الشرح بالعربي: لأن المستشفى عندها 2000 صورة فقط، فالبيانات قليلة لتدريب CNN عميقة من الصفر. الأفضل نستخدم Learning: Transfer نبدأ بنموذج CNN متدرب مسبقًا على ImageNet، ونعدّل الطبقة الأخيرة ونعمل Fine-tuning على صور الأشعة.

Reference: Lecture 5, Section 5.7 — Transfer learning; medical X-ray example

Part B – True or False

State whether each statement is true or false. For false statements, the correction is provided.

B1. Using a stride of 2 in a convolutional layer halves the size of the output.

Correct Answer: TRUE

Explanation (English): A stride of 2 moves the kernel two positions at a time and commonly reduces each spatial dimension to about half, depending on the input size, kernel and padding.

الشرح بالعربي: العبارة صحيحة في الحالة المعتادة: Stride = 2 يخلي الـ Kernel يتحرك بمقدار خطوتين، وبالتالي يقلل العرض والارتفاع تقريبًا للنصف. الحجم الدقيق يظل معتمدًا على الـ Kernel والـ Padding وحجم الإدخال.

Reference: Lecture 5, Section 5.2 — Stride and output size

B2. An RNN uses different weights W_x and W_h at each time step.

Correct Answer: FALSE

Correction: An RNN shares the same W_x and W_h weights across time steps.

Explanation (English): An RNN normally shares the same W_x and W_h across time steps; the same recurrent cell is applied repeatedly to the sequence.

الشرح بالعربي: العبارة خطأ. الـ RNN بتستخدم نفس مصفوفات الأوزان W_x وW_h في كل Time Step، وده اسمه Sharing Weight عبر الزمن، مش أوزان مختلفة لكل خطوة.

Reference: Lecture 5, Section 5.3 — Recurrent neural networks

B3. An LSTM has a separate cell state and three gates, each a small sigmoid layer giving values between 0 and 1.

Correct Answer: TRUE

Explanation (English): The standard LSTM has a cell state and three gates: input, forget and output. Their sigmoid activations produce values between 0 and 1.

الشرح بالعربي: العبارة صحيحة. الـ LSTM فيها State Cell وثلاث بوابات أساسية: Gate Input وGate Forget وGate Output. كل Gate بتستخدم Sigmoid، وبالتالي قيمتها بين 0 و1 لتتحكم في مرور المعلومات أو نسيانها.

Reference: Lecture 5, Section 5.4 — LSTM

B4. The transformer is built from attention, feed-forward layers and recurrence.

Correct Answer: FALSE

Correction: The transformer uses attention and feed-forward layers, with residual connections and positional encodings; it does not use recurrence.

Explanation (English): The transformer uses attention and feed-forward layers, along with residual connections and positional encodings; it does not use recurrence.

الشرح بالعربي: العبارة خطأ. الـ Transformer يعتمد على Attention وLayers Feed-Forward وConnections Residual وEncodings، Positional لكنه لا يعتمد على Recurrence مثل RNN.

Reference: Lecture 5, Section 5.5 — The transformer

B5. BERT is pretrained to predict the next token and is mainly used for generating text.

Correct Answer: FALSE

Correction: GPT predicts the next token; BERT is pretrained using masked-word prediction and is commonly used for language understanding.

Explanation (English): The lecture distinguishes BERT from GPT: GPT predicts the next token, while BERT is pretrained with a masked-word prediction objective and is commonly used for language understanding.

الشرح بالعربي: العبارة خطأ. توقع الـ Token التالي هو أسلوب GPT. أما BERT فيتعلم من خلال توقع الكلمات المحجوبة (Masked Language Modeling)، ويُستخدم غالبًا في مهام فهم اللغة.

Reference: Lecture 5, Section 5.5 — BERT and GPT

B6. In multi-head attention, each head has its own learned query, key and value matrices.

Correct Answer: TRUE

Explanation (English): Each attention head has its own learned projection matrices for queries, keys and values, allowing heads to learn different relationships.

الشرح بالعربي: العبارة صحيحة. كل Head Attention لها مصفوفات Query وKey وValue خاصة بها يتم تعلمها أثناء التدريب، وده بيسمح لكل Head بالتركيز على علاقات مختلفة بين الكلمات.

Reference: Lecture 5, Section 5.5 — Multi-head attention

B7. An autoencoder needs labelled examples, because each input must be paired with a class.

Correct Answer: FALSE

Correction: A standard autoencoder does not require class labels; it learns to reconstruct its input.

Explanation (English): A standard autoencoder is trained to reconstruct its input. The input itself supplies the reconstruction target, so class labels are not required.

الشرح بالعربي: العبارة خطأ. الـ Autoencoder لا تحتاج Class Labels، لأنها بتتعلم إعادة بناء نفس الـ Input، فالمدخل نفسه هو الهدف (Target) أثناء التدريب.

Reference: Lecture 5, Section 5.6 — Autoencoders

B8. Tiny, carefully chosen changes to an image that people cannot see can make a network confidently give a wrong answer.

Correct Answer: TRUE

Explanation (English): This describes adversarial examples: small, deliberately chosen perturbations can cause a neural network to make a confident incorrect prediction.

الشرح بالعربي: العبارة صحيحة. دي اسمها Examples: Adversarial تغييرات صغيرة ومحسوبة في الصورة، ممكن الإنسان ما يلاحظهاش، لكنها تخلي الشبكة تطلع توقعًا خاطئًا وبثقة عالية.

Reference: Lecture 5, Section 5.7 — Limitations: adversarial examples

B9. Max pooling increases the width and height of a feature map.

Correct Answer: FALSE

Correction: Max pooling downsamples the feature map and usually reduces its width and height.

Explanation (English): Max pooling downsamples a feature map. With 2×2 pooling and stride 2, it usually halves the width and height rather than increasing them.

الشرح بالعربي: العبارة خطأ. الـ Pooling Max بتقلل أبعاد الـ Feature Map، ومثال 2×2 مع Stride = 2 بيقلل العرض والارتفاع للنصف تقريبًا، مش بيزودهم.

Reference: Lecture 5, Section 5.2 — Pooling

B10. If all the training photos of cows contain grass, a network may learn “grass” instead of “cow” and fail on a cow on a beach.

Correct Answer: TRUE

Explanation (English): A network can learn a spurious correlation in the training data. It may associate grass with the cow label and fail when that background cue is absent.

الشرح بالعربي: العبارة صحيحة. الشبكة ممكن تتعلم ارتباطًا غير حقيقي (Spurious Pattern): بما إن كل صور الأبقار فيها عشب، ممكن تعتبر العشب علامة على وجود بقرة. لما تشوف بقرة على الشاطئ، ممكن تفشل في التعرف عليها.

Reference: Lecture 5, Section 5.7 — Limitations: spurious patterns

Final Answer Key

Part A: A1–C, A2–D, A3–B, A4–C, A5–D, A6–B, A7–D, A8–A, A9–D, A10–B, A11–A, A12–C, A13–D, A14–B, A15–A, A16–D, A17–B, A18–A, A19–C, A20–D, A21–B, A22–A, A23–D, A24–C, A25–B, A26–A, A27–C, A28–D, A29–A, A30–C

Part B: B1–T, B2–F, B3–T, B4–F, B5–F, B6–T, B7–F, B8–T, B9–F, B10–T